

Noetic Learning Math Contest 2026 Spring - Grade 8

Name: Aaron Parmar

1) What is 50% of 2026?

Correct! Answer:

2) The operation $\#$ is defined by $x \# y = x^2 - y^2$. What is the value of $11 \# 10$?

Correct! Answer:

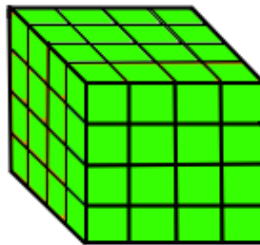
3) Jim, Sam, and Wade are on a basketball team. In the last game, the ratio of Jim's points to Sam's points to Wade's points was 3:7:5. Jim and Sam scored a total of 30 points. How many points did Wade score?



X Answer: points

Correct Answer:

4) A solid cube with side length 4 cm is painted green on all six faces and then cut into 1-cm cubes. How many of the 1-cm cubes have exactly two green faces?



X Answer: 1-cm cubes

Correct Answer:

5) What is the sum of the two largest prime factors of 2310?

✗ Answer:

Correct Answer:

6) Lucia and Maria live 1 mile from the gym. Lucia walks to the gym at a constant speed of 3 miles per hour, and Maria jogs to the gym at a constant speed of 5 miles per hour. They leave home at the same time. How many minutes sooner will Maria arrive at the gym than Lucia?

✗ Answer: minutes sooner

Correct Answer:

7) How many numbers between 100 and 200 are multiples of 3?

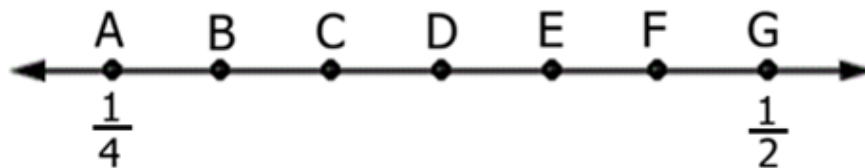
Correct! Answer: numbers

8) A fair six-sided die with faces numbered 1 through 6 is rolled twice. What is the probability that the product of the two numbers rolled is odd? Express your answer as a percent.

✗ Answer: %

Correct Answer:

9) On the number line below, points A through G are equally spaced. Point A represents $\frac{1}{4}$ and point G represents $\frac{1}{2}$. Which point represents $\frac{1}{3}$?



✗ Answer: Point

Correct Answer:

10) What is the ones digit of 27^{27} ?

✗ Answer:

Correct Answer:

11) Using the digits 4, 5, 7, 8, and 9, how many three-digit positive numbers divisible by 6 can be formed if no digit is repeated in each number?

✗ Answer: three-digit positive numbers

Correct Answer:

12) From the eight numbers below, how many different ways are there to choose six different numbers whose sum is 27? Note: The order of the numbers does not matter.



✗ Answer: different ways

Correct Answer:

13) In the addition problem below, the same letters represent the same digit, and different letters represent different digits. What digit does the letter A represent?

$$\begin{array}{r} A A \\ A B \\ + \quad A \\ \hline B A \end{array}$$

✗ Answer:

Correct Answer:

14) When the digits of a two-digit number are reversed, the new number is 36 less than the original number. In the original number, how much greater is the tens digit than the ones digit?

✗ Answer:

Correct Answer:

4

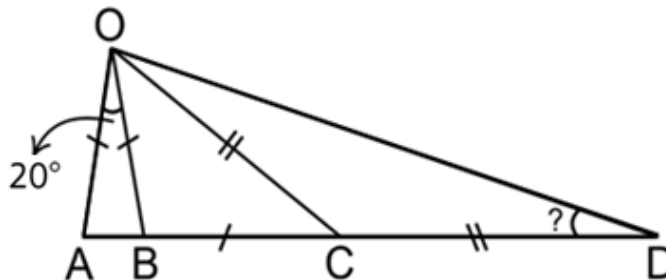
15) At Riverview Middle School, two-thirds of the students are in choir and one-half of the students are in band. Every student is in choir, band, or both. If 27 students are in both, what is the total number of students at Riverview Middle School?

✗ Answer: students

Correct Answer:

162

16) In triangle OAD , points B and C lie on segment AD , as shown below. $OA = OB = BC$ and $OC = CD$. The measure of $\angle AOB$ is 20° . What is the measure of $\angle CDO$?



✗ Answer: $^\circ$

Correct Answer:

20

17) It takes Ryan 6 hours to mow a soccer field by himself. It takes Seth 12 hours to mow the same soccer field by himself. Ryan starts mowing the field alone. Then, 5 hours later, Seth joins him. How many more minutes will it take them to finish mowing the field together? Note: Ryan and Seth each mow at a constant rate.

✗ Answer: more minutes

Correct Answer:

40

18) How many cups of an 18% alcohol solution should be added to 1 cup of a 30% alcohol solution to make a solution that is 20% alcohol?

✗ Answer: cups

Correct Answer:

5

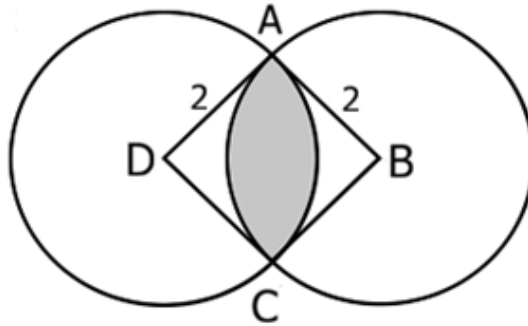
19) Daniela draws two acute triangles. Four of the six interior angles of the two triangles measure 80° , 75° , 40° , and 15° . What is the measure of the largest interior angle of the two triangles?

✗ Answer: $^\circ$

Correct Answer:

85

20) The side length of square ABCD is 2 inches. Two circles, each with a radius of 2 inches, are drawn with their centers at points B and D. What is the area of the region where the two circles overlap (the shaded region)? Use 3.14 for π .



✗ Answer: 3.14 square inches

Correct Answer:

2.28

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